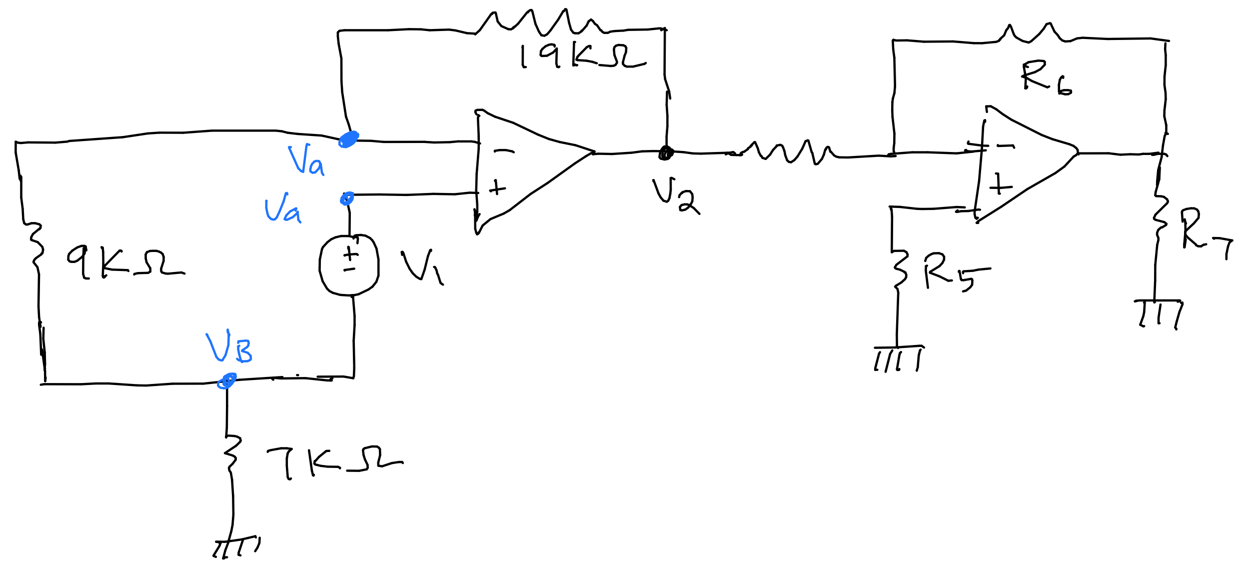
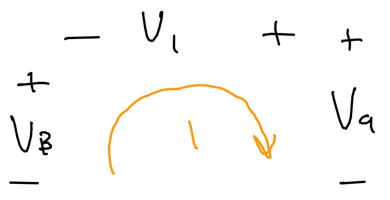




Applying KVL at loop 1

$$-V_B - V_1 + V_a = 0$$

$$V_a = V_B + V_1 \qquad (1)$$

Applying voltage division at loop 2

$$V_B = \frac{V_a \times 7}{7+9}$$

$$V_B = \frac{7V_a}{16} \qquad (2)$$

$$V_a = \frac{16V_B}{7}$$

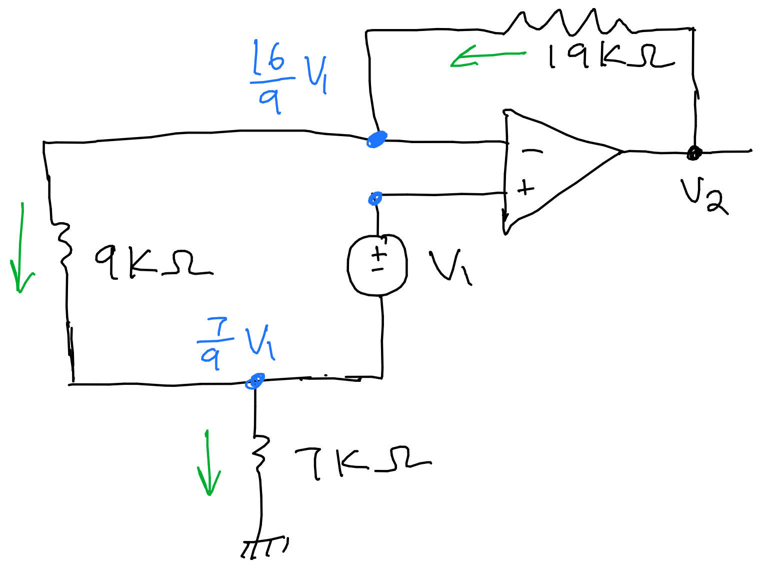
Substitute (2) into (1)

$$V_a = \frac{7}{16}V_a + V_1$$

$$V_a\left(1 - \frac{7}{16}\right) = V_1$$

$$V_a = \frac{16}{9}V_1$$

$$V_B = \frac{7V_a}{16} = \frac{7}{16} \times \frac{16}{9}V_1 = \frac{7}{9}V_1$$



$$\frac{V_2 - \frac{16}{9}V_1}{19000} = \frac{\frac{16}{9}V_1 - \frac{7}{9}V_1}{9000}$$

$$\frac{9}{19}V_2 - \frac{16}{19}V_1 = V_1$$

$$\frac{9}{19}V_2 = V_1\left(1 + \frac{16}{19}\right)$$

$$\frac{V_2}{V_1} = \frac{35}{19} \times \frac{19}{9}$$

$$\frac{V_2}{V_1} = \frac{35}{9} = 3{,}889$$